

Experiment: 9 Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective 1:

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

Objective 2:

Compare the signal-to-noise ratio for different communication scenarios using MATLAB. Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems.

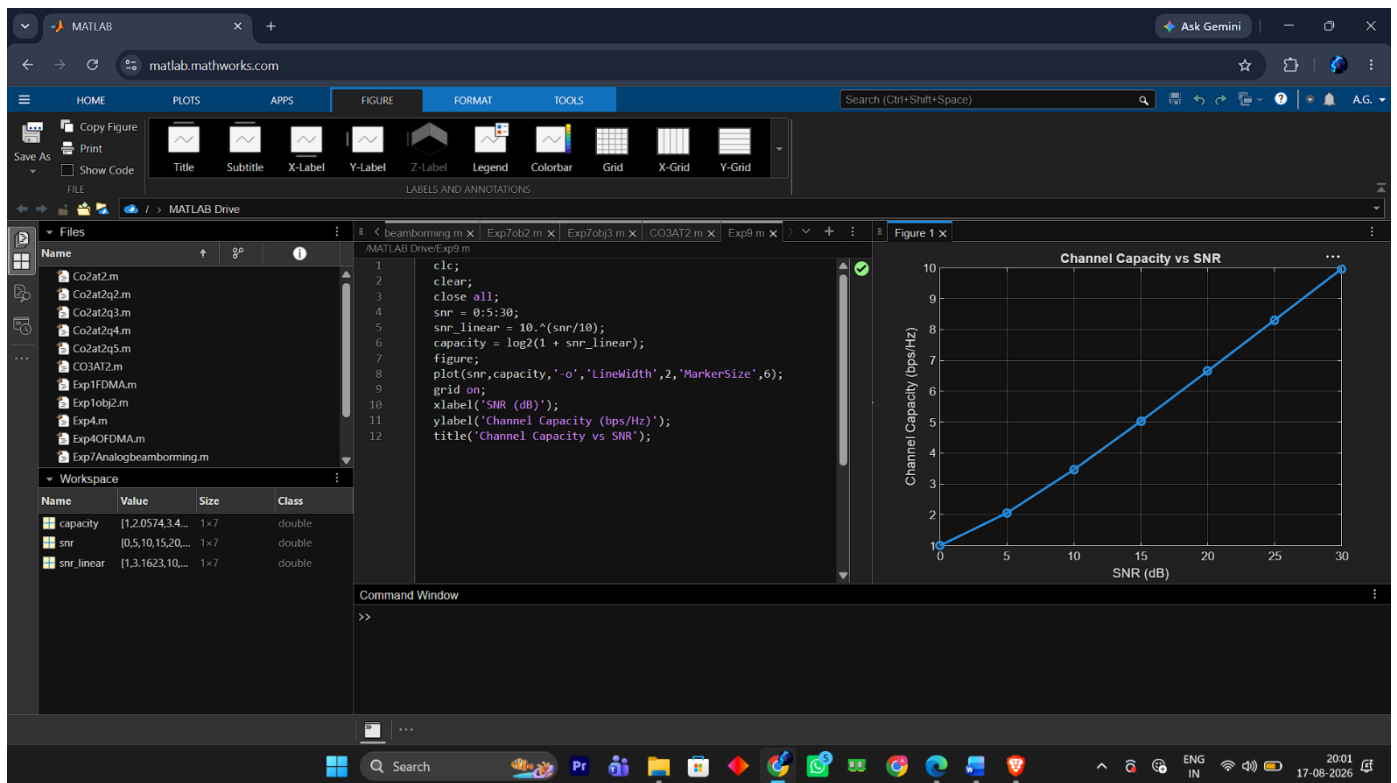
Objective 3:

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance.

Objective 1: Matlab Code and Output:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot (snr, capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel ('SNR (dB)');
ylabel ('Channel Capacity (bps/Hz)');
```

title ('Channel Capacity vs SNR');



Objective 2 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
snr = 0:5:30;
```

```
snr_linear = 10.^(snr/10);
```

```
capacity = log2(1 + snr_linear);
```

```
figure;
```

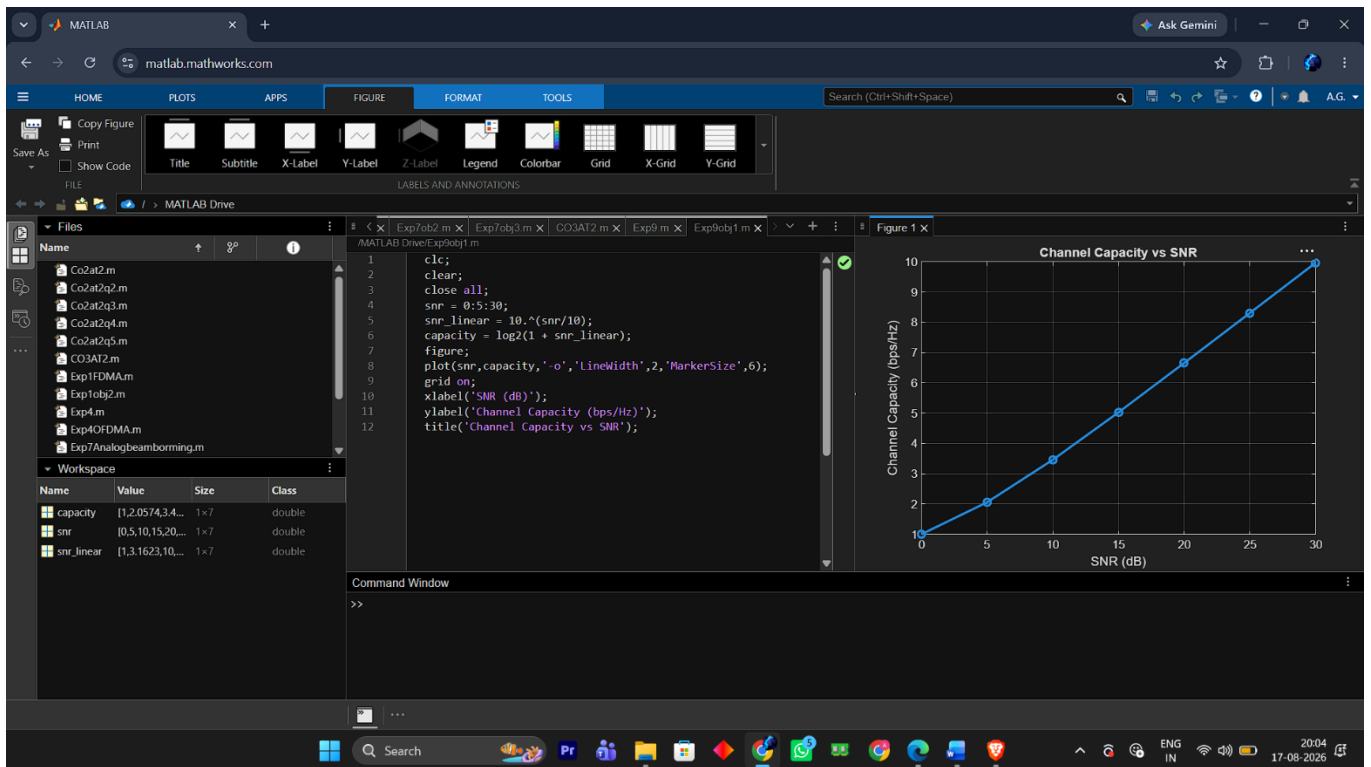
```
plot (snr, capacity,'-o','LineWidth',2,'MarkerSize',6);
```

```
grid on;
```

```
xlabel ('SNR (dB)');
```

```
ylabel ('Channel Capacity (bps/Hz)');
```

```
title ('Channel Capacity vs SNR');
```



Objective 3 Matlab Code and Output:

clc;

clear;

close all;

antennas = [4 8 16 32 64];

gain = 10*log10(antennas);

figure

bar (antennas, gain)

grid on

xlabel ('Number of Antenna Elements')

ylabel ('Antenna Gain (dB)')

title ('Antenna Gain for Different Antenna Array Sizes')

